Naushad Shaik Mohammad (Orcid ID: 0000-0001-8952-9581)

Pharmacogenetic profiling of dihydropyrimidine dehydrogenase (DPYD) variants in the Indian population

Shaik Mohammad Naushad[1], Tajamul Hussain[2,3], Salman Alrokayan[3,4], Vijay Kumar Kutala[5]

[1]Department of Pharmacogenomics, Sandor Speciality Diagnostics Pvt Ltd, Banjara Hills, Road No 3, Hyderabad; [2]Center of Excellence in Biotechnology Research, College of Science, King Saud University, Riyadh 11451, Saudi Arabia; [3]Research Chair for Biomedical Applications of Nanomaterials, Biochemistry Department, College of Science, King Saud University, Riyadh 11451, Saudi Arabia; [4]Biochemistry Department, College of Science, King Saud University, Riyadh 11451, Saudi Arabia, [5]Department of Clinical Pharmacology and Therapeutics, Nizam's Institute of Medical Sciences, Hyderabad, India.

Address for correspondence

Dr.Shaik Mohammad Naushad
Head, Biochemical Genetics & Pharmacogenomics
Sandor Speciality Diagnostics Pvt Ltd
Banjara Hills, Road No 3
Hyderabad-500034
Email: naushadsm@gmail.com
Ph: +91-7780446582

Running Title: DPYD variants in India

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1002/jgm.3289

Abstract

Background: To delineate the pharmacologically relevant dihydropyrimidine dehydrogenase (DPYD) variants in the Indian population.

Methods: We have screened 2000 Indian subjects for DPYD variants using Infinium Global Screening Array (GSA).

Results: The GSA analysis has identified seven coding, two intronic, and three synonymous DPYD variants. Level 1A alleles (rs75017182, rs3918290, P633Qfs*5, D949V) were found to be rare (minor allele frequency: 1.889%), while Level 3 alleles were observed to be predominant (C29R: 24.91%, I543V: 9.047%, M166V: 8.993%, V732I: 8.44%). *In silico* predictions have revealed that all Level 1A alleles were deleterious, while three (M166V, S534N, V732I) of seven Level 3 alleles were damaging. CUPSAT analysis has revealed that two Level 1A (P633Qfs*, D949V) and three Level 3 (I543V, V732I, S534N) variants were thermolabile. Pooled Indian data has showed that V732I, S534N, and rs3918290 variants are associated with 5-FU/Capecitabine toxicity, while C29R, I543V, and M166V variants exhibited the null association. A comparison of our data with the other population data from the "Allele Frequency Aggregator" database showed similarities with the South Asian data.

Conclusions: We have identified four Level 1A (non-functional/dysfunctional) and the seven Level 3 variants in the DPYD gene. Pooled Indian data has revealed the association of V732I, S534N, and rs3918290 variants with 5-FU/Capecitabine toxicity. Clustering analysis has shown the similarities in the DPYD profiles of the Indian and South Asian populations.

Keywords

Dihydropyrimidine dehydrogenase; 5-Fluorouracil; Capecitabine; adverse drug reactions; Global screening array.

Introduction

Dihydropyrimidine dehydrogenase (DPYD) catalyzes the reduction of uracil and thymine. 5-Fluorouracil (5-FU) is a prodrug, which undergoes phosphorylation to form 5-fluoro 2-deoxyuridine monophosphate (FdUMP) in the presence of thymidine phosphorylase (TP).[1] FdUMP is the active metabolite with the anti-cancer properties. DPYD facilitates the elimination of FU by reducing it to dihydrouracil (FUH_2), which converts to 5-Fluoroureidopropionic acid (FUPA) and 5-Fluoro-β-alanine.[1]

The DPYD deficiency (OMIM≠274270) might be asymptomatic or manifests as a convulsive disorder with motor and mental retardation.[2,3] Further, this deficiency can induce severe toxicity following 5-FU therapy in cancer patients.[4] The estimated incidence of partial DPYD deficiency was 3 to 8 % in the general population.[5] DPYD*2A (rs67376798), c.2846A>T, c.1679T>G and c.1236G>A variants contributed to increased FU toxicity in a large-scale study.[5] Genotype-guided dose adjustments of capecitabine and 5-FU prevented the toxicity.[5]

A Caucasian study (n=157) has identified 23 DPYD variants forming 15 haplotypes.[6] These haplotypes contributed to variable DPYD activity, which is correlated with the mRNA expression.[6] The DPYD deficiency was higher in African-Americans compared with Caucasians (8.0% vs. 2.8%).[7] A pooled Indian data showed that rs1801265 and rs120222243 variants are common in Indians.[8] Among 1064 Indian patients tested, 25.7% were heterozygous and 1.5% were homozygous for DPYD mutation.[9] However, this study did not describe the baseline characteristics of the subjects.[9] The frequencies of heterozygous (12.8% and 11.4%) and homozygous mutant (0.8% and 0.8%) genotypes of rs1801159 and rs1801160 variants were similar in this population.[9] The compound heterozygous genotype was present in 1.5% patients.[9] Reduced dose of FU was effective in minimizing the adverse drug events such as myelosuppression and mucositis in patients with DPYD mutations.[9] An Indian study has revealed 22 DPYD variants in 50 healthy adults, of which six variants were prevalent..[10]

In the present study, the distribution and clinical annotation of DPYD genetic variants in the Indian population (n=2000) was investigated. It provided a realistic estimate of the incidence of DPYD deficiency in India. Furthermore, the customized pharmacogenetic testing might help in optimizing the FU-mediated therapy.

Materials and Methods

Recruitment of subjects

We have recruited 2000 healthy adults (1215 men and 785 women) from July 2012 to December 2019. The age group of study subjects was 18 -72 yrs (mean ± SD: 45.9 ± 15.2 yrs). The mean body mass index was 25.9 ± 5.4 kgm^{-2}. None of the subjects had any history of malignancy and no family history related to cancer. Pre-recruitment blood glucose and blood pressure measurements were normal. None of them were on any medication. All the study participants consented to the study. **Table 1** describes the baseline characteristics of the studied population. The Institutional Ethical Committee of Sandor Proteomic Pvt Ltd, Hyderabad (EC/SRP/008/2012, dated 26.06.12) has approved the study protocol. We have collected whole blood samples in EDTA from these subjects.

Genotyping

We have extracted the genomic DNA from the whole blood using the standard protocol (QIAamp DNA mini kit, Qiagen, USA) and quantified the extracted DNA using the picogreen and nanodrop spectrophotometry. Infinium global screening array (GSA) v 1.0 carried the genetic analysis for 6,42,824 genetic markers in the following steps: whole genome amplification, fragmentation, hybridization, enzymatic base extension, and fluorescent staining. Illumina HiScan system facilitated the imaging of bead chips. Genome Studio V 2011.1 software performed the analysis and genotype calling.

The incorporation of sample-dependent and sample–independent controls at every stage of the experiment ensured quality control performance. Six-degree of freedom affine transformation algorithm normalized the allele-specific intensities. The performance of the experimental quality controls was satisfactory. The individuals with a missing genotype rate ≥0.03±2SD were excluded from the study. We have performed association QC per marker base and individual base.

Bioinformatics analysis

The Clinvar database provided information regarding each variant such as the nucleotide change, protein change, transcript details, frequency in other populations, and clinical relevance. The Mutalyzer module facilitated the information regarding nucleotide changes vs.protein changes by computing the standard Human Genome Sequence Variation Society (HGVS) nomenclature of each variant. The Mutation taster module predicted the disease potential using Bayes classifier. It was pretrained based on the data sets of > 390,000 known disease mutations (from HGMD) and > 6,800,000 harmless SNPs and Indel polymorphisms (from 1000 genome project). NNSplice, blastp and SwissProt predicted splicing effects, conservation of residues across the species, and protein features affected.

The Provean derived the average delta alignment score based on the top 30 identical clusters of sequences. The Prevean score distinguishes deleterious and neutral variants based on a cut-off value of -2.5. The SIFT module calculated the normalized probability of all possible substitutions through multiple sequence alignment. The deleterious mutations show a probability < 0.05 while neutral or tolerated mutations show a probability ≥ 0.05.

SNAP2, a trained neural network classifier, predicted the functional effect of mutations based on multiple sequence alignment, predicted secondary structure, solvent accessibility, and available annotation of close homologs. The regSNP-intron module predicted the disease-causing probability of intronic single nucleotide variants based on both genomic and protein structural features. The ENTPRISE-X module predicted the disease association of frameshift mutations based on the machine learning approach that measured Matthew's correlation coefficient.

The CUPSAT module calculated the mean force potentials of 40 atom types using inverse Boltzmann's principle. This module was pretrained based on a dataset of 4,024 non-redundant crystal structures of proteins using the PISCES algorithm. Further, the DSSP hydrogen bond estimation algorithm assigned the secondary structure to the amino acids of a protein and derived the torsion angles. Stepwise multiple regression predicted the thermal and denaturant stability by unifying the atom and torsion angle potentials.

Statistical analysis

The chi-square test verified whether the identified genotypes were distributed according to the Hardy-Weinberg equilibrium. Hierarchial clustering was performed using the ln(x+1) transformed data of population-specific frequencies of DPYD variants. All pairwise Euclidean (square root of sum of square) distances were calculated and the populations with the smallest distance were merged in each step. The heat map was generated using Euclidean distance and average linkage. Fisher's exact test assessed the association of DPYD variants with toxicity. Principal component analysis (PCA) differentiated the Level 1A and Level 3 variants using bioinformatics variables as inputs. This involved three step-process: i) Standardization of initial variables by transforming the data in ln (x+1) format; ii) Computation of covariance matrix; iii) Computation of the eigen vectors and eigen values of the covariant matrix to identify the principal components; iv) Feature Vector; and v) recasting the data along the principal components axes. The first principal component accounts for the largest possible variance in the given data set while the second principal component contributes to the next highest variance without correlating with the first principal component. Feature vector facilitates choice on which components to retain or discard based on their significance.

Results

Frequencies of DPYD variants

As shown in **Table 2**, we have identified 12 DPYD variants in our population. The rs1801265 was the most prevalent variant with a minor allele frequency (MAF) of 24.8%. The rs1801159, rs2297595, rs1801160, and rs17376848 variants showed the moderate prevalence with the MAFs of 9.05%, 9.00%, 8.38%, and 4.55% respectively. The rs56038477 and rs75017182 variants had MAFs 1.40% and 1.43%, respectively. The rs1801158, rs72549303, rs67376798, rs3918290 and rs3918289 variants are rare (MAF<1.0%). Out of 12 variants, seven were coding variants (C29R, I542V, M166V, V732I, S534N, P633Qfs*5, D949V), two were intronic (rs75017182 and rs3918290) and three were synonymous (rs17376848, rs56038477, rs3918289).

Bioinformatics analyses

As shown in **Table 3**, SIFT and Provean scores projected rs2297595, rs1801158, and rs67376798 as damaging. The rs72549303 is a frameshift mutation and hence, considered as damaging. The rs1801265, rs1801159, rs2297595, rs1801160, rs17376848, rs56038477, rs1801158, rs72549303, and rs3918289 variants were found to influence the splicing. The rs67376798 showed no splicing effect. The protein features such as FAD, FMN, [4Fe-4S] cluster binding differences were also explored.

As shown in **Table 4**, rs1801159, rs1801160, rs1801158, rs72549303, rs67376798 variants were predicted to be thermolabile. The protein stabilities of rs72549303 and rs67376798 variants were shown to be affected even in the presence of denaturant conditions.

Clinical annotation vs. Bioinformatics analyses

PharmaGKB assigns the clinical annotation levels of 1, 2, and 3 for variants with high, moderate, and low evidence. Level 4 is assigned for variants with preliminary studies. Clinical Pharmacogenetic Implementation Consortium (CPIC) or pharmacogenomics (PGx)-based annotation of variant-drug combination with the possible clinical recommendation categorized as Level 1A. Level 1 B is the annotation assigned to a variant-drug combination with a strong association in terms of highly significant ‘p’ values and with a strong effect size. Level 2A is the annotation assigned to a variant-drug combination where the variant is present in a very important pharmacogene as defined by PharmaGKB. Level 2B is the annotation assigned for a variant-drug combination with moderate evidence of association. Level 3 is the annotation assigned for a variant-drug combination based on a single significant study and lacking evidence of clear association in multiple studies.

There are four Level 1A variants, i.e. rs75017182, rs72549303, rs67376798, and rs3918290. The regSNP-intron module predicted higher disease probability scores for both the intronic variants, i.e. rs75017182 (score: 0.76) and rs3918290 (score: 0.80). The ENTPRISE-X module predicted deleterious nature of rs72549303 variant. The SNAP2, SIFT and Provean scores predicted the damaging or deleterious nature of the rs67376798 variant.

The SNAP2, SIFT and Provean scores predicted damaging or deleterious nature of three Level 3 variants (M166V, V732I, S534N). The bioinformatics tools differentiated the level 1A and Level 3 variants with 81.8% accuracy with a sensitivity of 100% and specificity of 71% as evidenced by the Principal component analysis (**Fig 1**).

Functional annotation vs. Bioinformatics analyses

The rs72549303 and rs3918290 non-functional variants were predicted to be deleterious and damaging by ENTPRISE X and regSNP-intron modules. The decreased function variants i.e. rs75017182 and rs67376798 variants were predicted to be disease-causing and damaging/deleterious by regSNP-intron, SNAP2, SIFT, and Provean scores. One decreased function variant rs56038477 although predicted to be neutral or tolerated by SIFT, and Provean scores, mutation taster depicted the splicing effects and the alteration in the protein features. All the functional variants were predicted to be neutral and tolerated by SIFT, and Provean Scores except rs2297595.

Comparison with other populations

Allele Frequency Aggregator (ALFA) database was used to retrieve population-specific frequencies of DPYD variants. This is a pipeline developed by National Center for Biotechnology Information (NCBI) based on 42 studies wherein 98,494 subjects were genotyped using exome analysis, whole genome, and SNP arrays. As shown in **Table 5**, among the variants with Level 1A significance, the population-specific frequencies corresponding to the rs72549303 variant were not available in the public domain.

For the MAFs of 11 DPYD variants compared, South Asian data showed no significant difference in MAFs compared to Indian data, except for the rs3918290 variant. The frequency of rs1801265 was the highest in African and African American populations. The frequencies of rs1801158, rs67376798, and rs3918290 variants were highest in European population. Indians showed higher frequency of rs1801265, rs2297595, rs1801160, rs75017182, and rs1801158 variants compared to other Asians. The frequencies of rs1801159 and rs17376848 were the highest in the Asian population. The DPYD clustered Indian, South Asian, and Asian populations as one group, African and African-American population as another group, and Europeans as a separate cluster. (**Fig 2**)

Clinical validation of association of DPYD variants in the pooled Indian data

As shown in **Table 6**, the pooling of Indian data of 5-FU/capecitabine toxicity[11-16] with the current study showed a lack of association of rs1801265, rs1801159, rs2297595 variants with FU-mediated toxicity. The rs1801160, rs1801158, and rs3918290 showed 1.88-, 4.40- and 158.14-fold increased risk for FU-mediated toxicity in Indians. As the rs3918290 showed heterogeneity in association with a broad 95% confidence interval, the studies related to only head and neck cancer (n=523) were explored further. Patients with high grade toxicity (Grade 3 to Grade 4) showed higher frequency of rs3918290 (5.59% vs. 1.57%) compared to the patients with lower grade toxicity (Grade 0 to Grade 2). Hence, this variant showed 3.79-folds (95% CI: 1.75 – 8.19, p=0.002) risk for 5-FU mediated toxicity in head and neck cancer. All three associations have 90% power with type 1 alpha error of 0.05.

Discussion

We report four DPYD variants of Level 1A significance and seven variants of Level 3 significance in the Indian population. The cumulative frequencies of Level 1A and Level 3

DPYD mutant alleles were 0.1889% and 58.31% . This is the first large-scale population study on DPYD from India. *In silico* predictions have revealed altered stability of rs72549303 and rs67376798 variants in both the thermal and denaturant experimental models. The intronic mutations predicted to increase disease probability encode a non-functional DPYD.[17,18] The rs3918290 intronic variant causes skipping for the entire exon, thus forming a non-functional protein.[17] The rs75017182 variant contributes to the partial production of a non-functional transcript by introducing a cryptic splice site.[18] The rs56038477 variant has a strong linkage disequilibrium with the rs75017182 variant.[18] These studies indicate that the four variants of Level 1A significance being either non-functional or have the decreased function.

Next, we compared our results with the largest cancer patient's data from India has revealed the similar distribution of rs1801159 (hetero: 16.9% vs. 14.28%; homo: 0.6% vs. 0.76%) and rs1801160 (hetero: 15.55% vs. 12.87%; homo: 0.6% vs. 0.75%).[9] Principal Component Analysis of bioinformatics data has differentiated the Level 1A and Level 3 variants.

The insights into the genotype versus biochemical phenotype of DPYD alleles were evident based on the following studies. The DPYD expression profiling in an isogenic mammalian system has revealed significantly higher activity in M166V and reduced or lack of activity in D949Y and P633Qfs*5 variants.[19] The C29R, I543V, V732I, and S534N variants showed higher DPYD activity.[20] The cloning and site-directed mutagenesis of human DPYD cDNA expressed in HEK293 Flp-In cells has revealed the DPYD activity profile of the major coding variants. D949V variant showed a 30% relative activity compared to the wild protein.[21] The activities of other variants were as follows: C29R (75%), M166V (72%), S534N (75%), I543V (90%), and V732I (*70%).[21]

A study on 603 cancer patients tested the clinical validity of eight DPYD variants in predicting FU-mediated toxicity.[22] The rs67376798, rs3918290 and rs55886062 variants emerged as the most important pharmacogenetic determinants of FU-mediated Grade ≥ 3 toxicity after the bootstrap validation and Bonferroni's correction. The reduction of FU dose has prevented such toxicity in DPYD mutants.[22] A study on 508 colon cancer patients has revealed a higher incidence of fluoropyrimidine mediated adverse events in rs1801160 A-allele carriers and rs2297595 GG-genotype carriers.[23] The rs1801160 and rs3918290 variant alleles showed increased risk for shorter time to neutropenia.[19]

A few Indian studies have explored the association of DPYD variants with 5-FU/capecitabine-mediated toxicity.[11-16, 24] Low DPYD expression was linked to unfavourable overall survival in colorectal cancer patients.[24] DPYD*9A (rs1801265) variant has found to increase the risk for the hand-foot syndrome, diarrhea, and thrombocytopenia following treatment with capecitabine and oxaliplatin in colorectal cancer.[11] Carriers of rs1801265, rs1801159, rs1801160 variants have exhibited severe mucositis and severe diarrhea following 5-FU therapy in head and neck cancer.[12] DPYD deficiency is the major contributor of high grade toxicity in cancer accounting for 70.96% cases.[13] The rs1801265, rs1801159, rs2297595, rs1801160, and rs3918290 variants were responsible for the high grade toxicity in these cancer patients.[13] The frequencies of rs3918290 and rs1801158 were higher in non-responders to cisplatin and 5-FU compared to responders in head and neck cancer.[14] Two patients heterozygous for rs3918290 showed the high grade toxicity following combination therapy of 5-FU and cisplatin in head and neck cancer (n=23).[15] Breast cancer patients harboring the rs3918290 variant showed poor response to 5-FU.[16]

Above mentioned Indian studies have demonstrated the association of rs1801160 and rs1801158 variants with FU/Capecitabine-mediated toxicity. This could be due to the predicted thermolabile nature of these variants . The rs3918290 variant is a documented risk factor for FU/Capecitabine-mediated toxicity. The functional annotation of variant alleles facilitates the derivation of activity scores thus categorizing them based on metabolizer phenotypes.[25] Absence of any DPYD variant alleles or presence of functional variants i.e. rs1801265, rs1801159, and rs2297595 variants suggests a normal metabolizer phenotype. CPIC has recommended a standard dose of 5-FU or fluoropyrimidine in normal metabolizers, a dose reduction in intermediate metabolizers, and alternate therapy in poor metabolizers.[25]

The major strengths of the current study are: i) large-scale population-based evaluation of DPYD variants using GSA; ii) application of bioinformatics variables in the segregation of DPYD variant alleles into Level 1A and Level 3 alleles using Principal Component Analysis; iii) clustering of DPYD genotype data to identify the populations that are similar in genetic architecture; iv) pooled analysis of Indian data to assess the pharmacological relevance of DPYD variants in 5-FU/Capecitabine mediated toxicity. To date, six DPYD variants were clinically annotated from India. Future studies should explore the relevance of other variants including the rare variants (MAFs<1.0%).

The current study facilitates the implementation of a pharmacogenetic panel comprising 12 variants (four of Level 1A and seven of Level 3 significance) in Indian patients and justifies the strong association of two variants by pooling the existing Indian data. It utilizes bioinformatics tools for clinical and functional annotation of these variants by considering the impact of genetic variation on the protein function and its stability.

To conclude, this population-based study has evaluated four DPYD variants of Level 1A significance and seven DPYD variants of Level 3 significance. Thus, this study laid the foundation for future studies by providing the baseline data on the distribution of these variants. Pooled Indian data showed the association of rs1801160, rs1801158, and rs3918290 variants with 5-FU/Capecitabine-mediated toxicity. The functional relevance of these variants was deduced using regSNP-intron, ENTPRISE-X, SNAP2, SIFT, Provean, Mutalyzer, Mutation Taster, CUPSAT modules. Indian population has showed the genetic cluster matching with the South Asians, while African/African Americans and Europeans formed the separate clusters.

Acknowledgments

Authorship: The study was conceived and designed by SMN and VKK. The recruitment of the subjects, genetic and bioinformatics analyses were performed by SMN, TH, SAA, and VKK. The data was interpreted and the manuscript was drafted by SMN and VKK. The manuscript was revised for the critical content by TH and SAA.

Funding: The study was funded by intramural grants of Sandor Speciality Diagnostics Pvt Ltd, Hyderabad, India. The authors are grateful to the Deanship of Scientific Research, King Saud University for funding through the Vice Deanship of Scientific Research Chairs.

Conflicts of Interest Statement

All the authors hereby declare no conflicts of interest.

Data availability Statement

The data will be made available on request.

References

1. Büchel B, Rhyn P, Schürch S, Bühr C, Amstutz U, Largiadèr CR. LC-MS/MS method for simultaneous analysis of uracil, 5,6-dihydrouracil, 5-fluorouracil and 5-fluoro-5,6-dihydrouracil in human plasma for therapeutic drug monitoring and toxicity prediction in cancer patients. Biomed Chromatogr. 2013 Jan;27(1):7-16.

2. Chen BC, Mohd Rawi R, Meinsma R, Meijer J, Hennekam RC, van Kuilenburg AB. Dihydropyrimidine dehydrogenase deficiency in two Malaysian siblings with abnormal MRI findings. Mol Syndromol. 2014 Dec;5(6):299-303.

3. Fleger M, Willomitzer J, Meinsma R, Alders M, Meijer J, Hennekam RCM, Huemer M, van Kuilenburg ABP. Dihydropyrimidine Dehydrogenase Deficiency: Metabolic Disease or Biochemical Phenotype? JIMD Rep. 2017;37:49-54.

4. Papanastasopoulos P, Stebbing J. Molecular basis of 5-fluorouracil-related toxicity: lessons from clinical practice. Anticancer Res. 2014 Apr;34(4):1531-5.

5. Henricks LM, Opdam FL, Beijnen JH, Cats A, Schellens JHM. DPYD genotype-guided dose individualization to improve patient safety of fluoropyrimidine therapy: call for a drug label update. Ann Oncol. 2017 Dec 1;28(12):2915-2922.

6. Seck K, Riemer S, Kates R, Ullrich T, Lutz V, Harbeck N, Schmitt M, Kiechle M, Diasio R, Gross E. Analysis of the DPYD gene implicated in 5-fluorouracil catabolism in a cohort of Caucasian individuals. Clin Cancer Res. 2005 Aug 15;11(16):5886-92.

7. Mattison LK, Fourie J, Desmond RA, Modak A, Saif MW, Diasio RB. Increased prevalence of dihydropyrimidine dehydrogenase deficiency in African-Americans compared with Caucasians. Clin Cancer Res. 2006 Sep 15;12(18):5491-5.

8. Hariprakash JM, Vellarikkal SK, Keechilat P, Verma A, Jayarajan R, Dixit V, Ravi R, Senthivel V, Kumar A, Sehgal P, Sonakar AK, Ambawat S, Giri AK, Philip A, Sivadas A, Faruq M, Bharadwaj D, Sivasubbu S, Scaria V. Pharmacogenetic landscape of DPYD variants in south Asian populations by integration of genome-scale data. Pharmacogenomics. 2018 Feb;19(3):227-241.

9. Patil VM, Chougule A, Noronha V, Joshi A, Abhyankar A, Ostwal V, Ramaswamy A, Bagayatkar P, Banavali S, Prabhash K. DPYD Mutation in Indian Patients. Clin Oncol (R Coll Radiol). 2019 Oct;31(10):732-733.

10. Iyer SN, Singhal RS, Hegde MR, Ankala A. Genetic variation in dihydropyrimidine dehydrogenase (DPYD) gene in a healthy adult Indian population. Ann Hum Biol. 2015 Jan;42(1):97-100.

11. Varma K A, Jayanthi M, Dubashi B, Shewade DG. Influence of DPYD*9A, DPYD*6 and GSTP1 ile105val Genetic Polymorphisms on Capecitabine and Oxaliplatin (CAPOX) Associated Toxicities in Colorectal Cancer (CRC) Patients. Asian Pac J Cancer Prev. 2019 Oct 1;20(10):3093-3100.

12. Patil VM, Noronha V, Joshi A, Zanwar S, Ramaswamy A, Arya S, Mahajan A, Bhattacharjee A, Prabhash K. Dihydropyrimidine dehydrogenase mutation in neoadjuvant chemotherapy in head and neck cancers: Myth or reality? South Asian J Cancer. 2016 Oct-Dec;5(4):182-185.

13. Sahu A, Ramaswamy A, Ostwal V. Dihydro pyrimidine dehydrogenase deficiency in patients treated with capecitabine based regimens: a tertiary care centre experience. J Gastrointest Oncol. 2016 Jun;7(3):380-6.

14. Dhawan A, Ruwali M, Pant MC, Rahman Q, Parmar D. Association of genetic variability in enzymes metabolizing chemotherapeutic agents with treatment response in head and neck cancer cases. Asia Pac J Clin Oncol. 2017 Apr;13(2):e11-e20.

15. Dhawan D, Panchal H, Shukla S, Padh H. Genetic variability & chemotoxicity of 5-fluorouracil & cisplatin in head & neck cancer patients: a preliminary study. Indian J Med Res. 2013 Jan;137(1):125-9.

16. Ch. Kalyana Kumar, Sudha Murthy and Kaiser Jamil. Possible Associations of Splice Site Mutation of Dihydropyrimidine Dehydrogenase (IVS14+1G>A) in Adverse Drug Reactions in Some Invasive Ductal Carcinoma Patients. Inter J Pharm. 2007;3: 130-136.

17. Van Kuilenburg AB, Vreken P, Beex LV, Meinsma R, Van Lenthe H, De Abreu RA, van Gennip AH. Heterozygosity for a point mutation in an invariant splice donor site of dihydropyrimidine dehydrogenase and severe 5-fluorouracil related toxicity. Eur J Cancer. 1997 Nov;33(13):2258-64.

18. Nie Q, Shrestha S, Tapper EE, Trogstad-Isaacson CS, Bouchonville KJ, Lee AM, Wu R, Jerde CR, Wang Z, Kubica PA, Offer SM, Diasio RB. Quantitative Contribution of rs75017182 to Dihydropyrimidine Dehydrogenase mRNA Splicing and Enzyme Activity. Clin Pharmacol Ther. 2017 Oct;102(4):662-670.

19. Offer SM, Fossum CC, Wegner NJ, Stuflesser AJ, Butterfield GL, Diasio RB. Comparative functional analysis of DPYD variants of potential clinical relevance to dihydropyrimidine dehydrogenase activity. Cancer Res. 2014 May 1;74(9):2545-54.

20. Offer SM, Wegner NJ, Fossum C, Wang K, Diasio RB. Phenotypic profiling of DPYD variations relevant to 5-fluorouracil sensitivity using real-time cellular analysis and in vitro measurement of enzyme activity. Cancer Res. 2013 Mar 15;73(6):1958-68.

21. Kuilenburg ABPV, Meijer J, Tanck MWT, Dobritzsch D, Zoetekouw L, Dekkers LL, Roelofsen J, Meinsma R, Wymenga M, Kulik W, Büchel B, Hennekam RCM, Largiadèr CR. Phenotypic and clinical implications of variants in the dihydropyrimidine dehydrogenase gene. Biochim Biophys Acta. 2016 Apr;1862(4):754-762.

22. Toffoli G, Giodini L, Buonadonna A, Berretta M, De Paoli A, Scalone S, Miolo G, Mini E, Nobili S, Lonardi S, Pella N, Lo Re G, Montico M, Roncato R, Dreussi E, Gagno S, Cecchin E. Clinical validity of a DPYD-based pharmacogenetic test to predict severe toxicity to fluoropyrimidines. Int J Cancer. 2015 Dec 15;137(12):2971-80.

23. Ruzzo A, Graziano F, Galli F, Galli F, Rulli E, Lonardi S, Ronzoni M, Massidda B, Zagonel V, Pella N, Mucciarini C, Labianca R, Ionta MT, Bagaloni I, Veltri E, Sozzi P, Barni S, Ricci V, Foltran L, Nicolini M, Biondi E, Bramati A, Turci D, Lazzarelli S, Verusio C, Bergamo F, Sobrero A, Frontini L, Menghi M, Magnani M. Dihydropyrimidine dehydrogenase pharmacogenetics for predicting fluoropyrimidine-

related toxicity in the randomised, phase III adjuvant TOSCA trial in high-risk colon cancer patients. Br J Cancer. 2017 Oct 24;117(9):1269-1277.

24. Gajjar KK, Vora HH, Kobawala TP, Trivedi TI, Ghosh NR. Deciphering the potential value of 5-fluorouracil metabolic enzymes in predicting prognosis and treatment response of colorectal cancer patients. Int J Biol Markers. 2018 May;33(2):180-188.

25. Amstutz U, Henricks LM, Offer SM, Barbarino J, Schellens JHM, Swen JJ, Klein TE, McLeod HL, Caudle KE, Diasio RB, Schwab M. Clinical Pharmacogenetics Implementation Consortium (CPIC) Guideline for Dihydropyrimidine Dehydrogenase Genotype and Fluoropyrimidine Dosing: 2017 Update. Clin Pharmacol Ther. 2018 Feb;103(2):210-216.

Table 1. Baseline characteristics of the studied population

Parameter	**Mean or proportion**
Age (mean±SD) yr	45.9±15.2
Males:Females	1215: 785
Body mass index in Kg/m^2	25.9±5.4
Family history of cancer	0/2000
Diabetes	0/2000
Hypertension	0/2000
Smoking	115/2000 (5.75%)
Alcohol intake	219/2000 (10.95%)

Table 2. Distribution of DPYD variants in the Indian population

rs number	**Nucleotide change**	**Protein change**	WW	WM	MM	**MAF (%)**	**P_{HWE}**
rs1801265 (*9A)	c.85 T>C	C29R	1128	752	120	24.80	0.94
rs1801159 (*5)	c.1627 A>G	I543V	1650	338	12	9.05	0.42
rs2297595	c.496 A>G	M166V	1655	330	15	9.00	0.96
rs1801160 (*6)	c.2194 G>A	V732I	1677	311	12	8.38	0.84
rs17376848	c.1896 T>C	F632=	1821	176	3	4.55	0.87
rs56038477	c.1236 G>A	E412=	1944	56	0	1.40	1.00
rs75017182 (HapB3)	c.1129-5923C>G	Intron 10	1943	57	0	1.43	1.00
rs1801158	c.1601G>A	S534N	1965	35	0	0.88	1.00
rs72549303	c.1898delC	P633Qfs*5	1996	4	0	0.10	1.00
rs67376798	c.2846A>T	D949V	1997	3	0	0.075	1.00
rs3918290 (*2A)	c.1905+1G>A	IVS14+1G>A	1998	2	0	0.05	1.00
rs3918289	c.1905C>T	N635=	1999	1	0	0.025	1.00

WW: wild; WM: heterozygous; MM: homozygous mutant; MAF: Minor allele frequency; HWE: Hardy-Weinberg equilibrium.

Table 3. Bioinformatics analysis

rs number	Level	Function	regSNP-intron	ENTPRISEX	SNAP2	SIFT	Provean	Splicing effect	FAD	FMN	(4Fe-4S] cluster
rs1801265	3	Normal (NM)			22 (weak)	Tolerated (0.197)	Neutral (0.08)	GTGTTCCACTTC/ GGCC	Yes	Yes	Yes
rs1801159	3	Normal (NM)			10 (weak)	Tolerated (0.111)	Neutral (0.73)	TATAAATCCTTTT/ GGTC	No	Yes	Yes
rs2297595	3	Normal (NM)			60 (strong)	Damaging (0.004)	Deleterious (-3.50)	GTAT/ TCAAAGCAA	Yes	Yes	Yes
rs1801160	3	Normal (NM)			65 (strong)	Tolerated (0.057)	Neutral (-0.98)	GTTACA/ GCCA	No	No	No
rs17376848	3	Normal (NM)			ND	Tolerated (1.000)	Neutral (0.00)	TCCA/ GACA	No	No	No
rs56038477	3	Decreased (IM)			ND	Tolerated (1.000)	Neutral (0.00)	GAGCAA/ GATG	Yes	Yes	Yes
rs75017182	1A	Decreased (IM)	Disease Probability (0.76)		ND	ND	ND	ND	ND	ND	ND
rs1801158	3	Normal (NM)			71 (strong)	Damaging (0.007)	Deleterious (-2.72)	GACA/ TTAG	No	Yes	Yes
rs72549303	1A	No function (PM)		Deleterious (0.61)	ND	ND	ND	CCAG/ ACAA	No	Yes	Yes
rs67376798	1A	Decreased (IM)			86 (strong)	Damaging (0.000)	Deleterious (-7.16)	No effect	No	No	Yes
rs3918290	1A	No function (PM)	0.80		ND	Damaging (0.002)	Neutral (-1.77)	ND	ND	ND	ND
rs3918289	--	Normal			ND	Tolerated (0.313)	Neutral (0.00)	CAAC/ GTAA	No	Yes	Yes

Level 1A: Clinical Pharmacogenetic Implementation Consortium (CPIC) or pharmacogenomics (PGx)-based annotation of variant-drug combination with the possible clinical recommendation; Level 3: CPIC annotation assigned for a variant-drug combination based on a single significant study and lacking evidence of clear association in multiple studies; NM: Normal metabolizer; IM: Intermediate metabolizer; PM: Poor metabolizer; FAD; Flavin adenine dinucleotide; FMN: Flavin mononucleotide; [4Fe-4S] cluster: iron-sulphur cluster; ND: not determined.

Note: The information of Level and Function was obtained by referring PharmaGKB and CPIC databases.

Table 4. Protein stability and torsion assessment based on CUPSAT analysis

rs number	Protein change	Thermal			Denaturant		
		Overall Stable	**Torsion**	**Predicted ΔΔG (Kcal/mol)**	**Overall Stable**	**Torsion**	**Predicted ΔΔG (Kcal/mol)**
rs1801265	C29R	Yes	Unfavourable	0.53	Yes	Unfavourable	0.40
rs1801159	I543V	No	Unfavourable	-0.53	Yes	Favourable	1.51
rs2297595	M166V	Yes	Unfavourable	0.50	Yes	Unfavourable	0.62
rs1801160	V732I	No	Unfavourable	-3.26	Yes	Unfavourable	0.63
rs1801158	S534N	No	Unfavourable	-1.3	Yes	Unfavourable	3.16
rs67376798	D949V	No	Favourable	-0.33	No	Favourable	-0.19

ΔΔG: change in Gibb's free energy

Table 5. The distribution of DPYD variants across different populations

Variant	Indian		South Asian		Asian		European		African		African American	
	M/W	OR (95% CI)	M/W	OR (95% CI)	M/W	OR (95% CI)	M/W	OR (95% CI)	M/W	OR (95% CI)	M/W	OR (95% CI)
rs1801265	992/ 3008	Ref	2550/ 7242	1.07 (0.98 - 1.16)	48/ 792	0.18 (0.14 – 0.25)*	67662/ 242782	0.85 (0.79 – 0.91)*	5352/ 7936	2.05 (1.89 – 2.22)*	5178/ 7678	2.05 (1.89 – 2.22)*
rs1801159	362/ 3638	Ref	860/ 8948	0.97 (0.85 – 1.10)	220/ 628	3.52 (2.92 – 4.25)*	61804/ 247012	2.52 (2.26 – 2.80)*	2024/ 11232	1.81 (1.61 – 2.04)*	1954/ 10874	1.81 (1.60 – 2.03)*
rs2297595	360/ 3640	Ref	910/ 8902	1.03 (0.91 – 1.18)	12/ 840	0.14 (0.08 – 0.26)*	30898/ 288586	1.08 (0.97 – 1.20)	470/ 12846	0.37 (0.32 – 0.43)*	464/ 12420	0.38 (0.33 – 0.44)*
rs1801160	335/ 3665	Ref	170/ 1786	1.04 (0.86 – 1.26)	10/ 330	0.33 (0.18 – 0.63)*	9079/ 162117	0.61 (0.55 – 0.69)*	110/ 1578	0.76 (0.61 – 0.95)*	109/ 1547	0.77 (0.62 – 0.97)
rs17376848	182/ 3818	Ref	354/ 9446	0.79 (0.66 – 0.94)	112/ 596	3.94 (3.07 – 5.07)*	11462/ 258690	0.93 (0.80 – 1.08)	186/ 7722	0.51 (0.41 – 0.62)*	178/ 7466	0.50 (0.41 0 0.62)*
rs56038477	56/ 3944	Ref	37/ 1919	1.36 (0.89 – 2.06)	0/188	0.00	1182/ 61862	1.35 (1.03 – 1.76)*	10/ 1470	0.48 (0.24 – 0.94)*	10/ 1438	0.49 (0.25 – 0.96)*
rs75017182	57/ 3943	Ref	37/ 1919	1.33 (0.88- 2.03)	7/ 16629	0.03 (0.01 – 0.06)*	48/ 1964	1.69 (1.15 – 2.49)	2/ 2642	0.05 (0.01 – 0.22)*	246/ 64782	0.26 (0.20 – 0.35)*
rs1801158	35/ 3965	Ref	10/ 1946	0.58 (0.29 – 1.18)	286/ 49906	0.65 (0.46 – 0.92)*	3825/ 172923	2.51 (1.79 – 3.50)*	19/ 1833	1.17 (0.67 – 2.06)	18/ 1794	1.14 (0.64 – 2.01)
rs67376798	3/ 3997	Ref	2/ 1954	1.22 (0.22- 2.53)	18/ 50282	0.48 (0.14 – 1.62)	68/ 16200	5.59 (1.76- 17.78)	14/ 20786	0.90 (0.26 – 3.12)	102/ 64930	2.09 (0.66 – 6.60)
rs3918290	2/ 3998	Ref	34/ 9762	6.96 (1.67 – 28.99)*	0/ 16636	0.00	1108/ 230560	9.61 (2.40 – 38.47)*	5/ 17423	0.88 (0.37 – 1.17)	60/ 64972	1.85 (0.45 – 7.56)
rs3918289	1/ 3999	Ref	2/ 11510	0.70 (0.06 – 7.67)	2/ 16634	0.48 (0.04 – 5.30)	108/ 249188	1.73 (0.24 – 12.42)	2/ 7594	1.05 (0.10 – 11.62)	2/ 7338	1.09 (0.10 – 12.02)
rs72549303	4/ 3996	Ref	--	---	--	---	--	---	--	---	---	---

M: Mutant alleles; W: wild alleles; OR: unadjusted odds ratio; CI: confidence interval; The data related to other population was obtained from the Allele Frequency Aggregator (ALFA) database.

Table 6. Distribution of DPYD variants in the Indian population

rs number	**Current study**		**Patil et al[12]**		**Sahu et al[13]**		**Dhawan et al[14]**		**Varma et al[11]**		**Kalyan Kumar et al[16]**		**ADR data (cumulative)**		**OR**	**95% CI**	**P**
	W	M	W	M	W	M	W	M	W	M	W	M	W	M			
rs1801265	3008	992	19	5	32	12	-	-	46	16			97	33	1.03	0.69-1.54	0.95
rs1801159	3638	362	22	2	36	8	-	-	-	-			58	10	1.73	0.88-3.42	0.18
rs2297595	3640	360	21	3	38	6	-	-	-	-			59	9	1.54	0.76-3.14	0.32
rs1801160	3666	334	17	7	40	4	-	-	54	8			111	19	1.88	1.14-3.10	0.03
rs1801158	3962	38	23	1	-	-	320	10	-	-			498	21	4.40	2.56 – 7.55	<0.0001
rs3918290	3998	2	-	-	28	16	251	13	-	-	214	10	493	39	158.14	38.07 - 656.89	<0.0001

M: male; F: female; Age expressed as mean ± SD yr; MAF: Minor allele frequency; OR: unadjusted odds ratio; CI: confidence interval.

Basic characteristics of the Indian studies enrolled:

Study name	Cancer type	M:F ratio	Age group (yr)	Genotyping method
Patil et al	Head and neck	25:9	43±21	Sanger
Sahu et al	Colorectal, stomach and gall blader cancer	22:9	56±22	Sanger
Dhawan et al	Head and neck cancer	500:0	……	RFLP
Varma et al	Colon cancer	90:55	50±13	Taqman
Kalyan Kumar et al	Breast cancer	0:112	25 - 68	RFLP

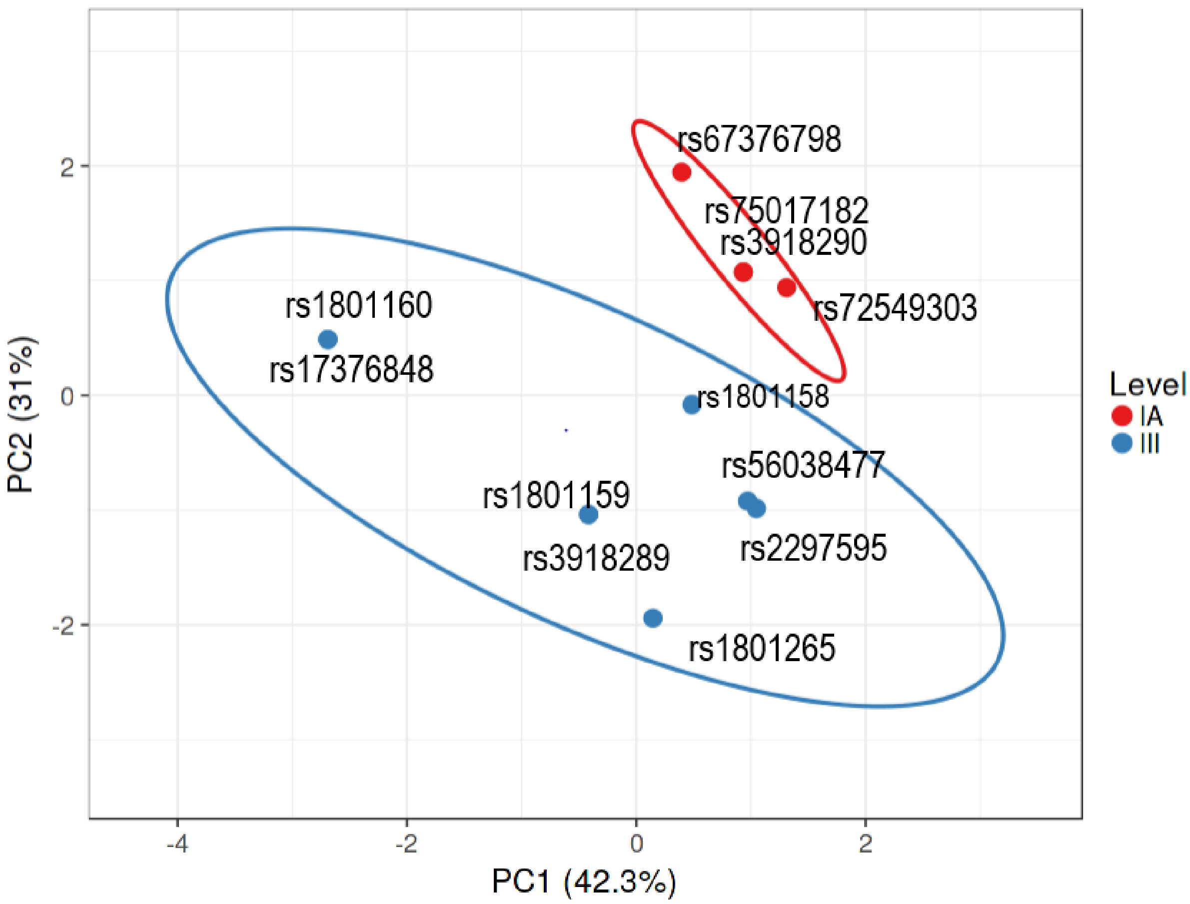

Fig 1. Principal component analysis differentiation Level I and Level III variables

The bioinformatics data comprising of impact of variation on the protein function and protein features were used as inputs. Unit variance scaling is applied to rows; SVD with imputation is used to calculate principal components. X and Y axis show principal component 1 and principal component 2 that explain 42.3% and 31% of the total variance, respectively. Prediction ellipses are such that with probability 0.95, a new observation from the same group will fall inside the ellipse. N = 12 data points.

Level 1A: Clinical Pharmacogenetic Implementation Consortium (CPIC) or pharmacogenomics (PGx)-based annotation of variant-drug combination with the possible clinical recommendation. Level 3: CPIC annotation assigned for a variant-drug combination based on a single significant study and lacking evidence of clear association in multiple studies.

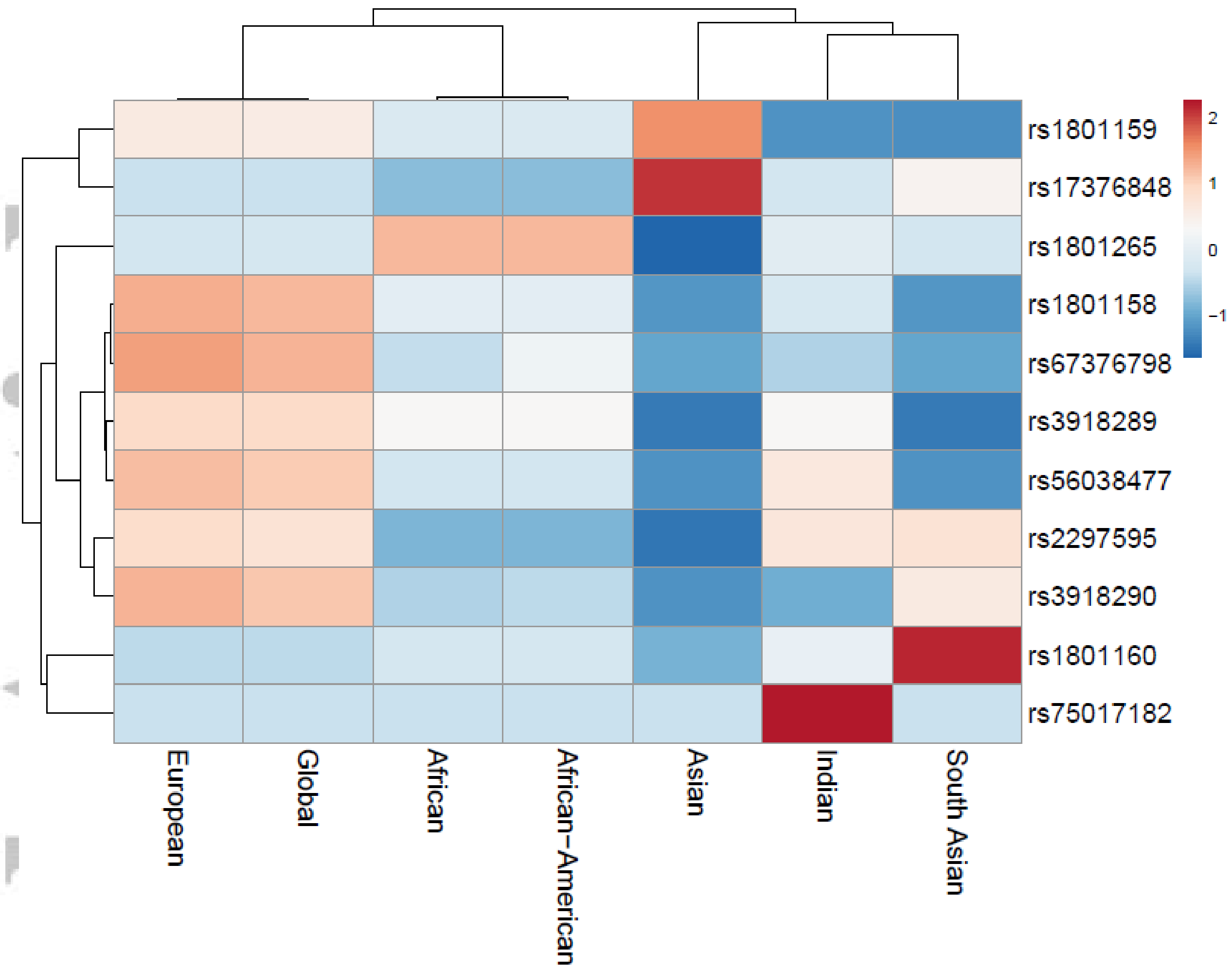

Fig 2. Heat map analysis of DPYD genotype based gene clusters across different populations

This illustrates the distribution of 11 DPYD variants six populations and global population. Indian data clustered with the data of South Asians and Asians. Africans and African-Americans were clustered together. Europeans formed separate cluster. Euclidean distances and linkage statistics were used for this clustering on ln(x+1) transformed data.

Note: The minor allele frequencies data was transformed into ln (X+1) format and depicted in the range of -1 to 2. -1 indicates lowest frequency while 2 indicates highest frequency.

Graphical Abstract Text

Population study using Global Screening Array (GSA):

GSA analysis in 2000 subjects revealed 12 DPYD variants: seven coding variants (C29R, I542V, M166V, V732I, S534N, P633Qfs*5, D949V); two intronic (rs75017182 and rs3918290); and three were synonymous (rs17376848, rs56038477, rs3918289). The minor allele frequency (MAF) of Level 1A (rs75017182, rs3918290, P633Qfs*5, D949V) variant alleles was 1.6814%. The MAFs of Level 3 variants were: C29R (MAF: 24.91%), I543V (MAF: 9.047%), M166V (8.993%), V732I (8.44%).

Bioinformatics analysis:

M166V, V732I, D949V are predicted to be damaging. Thermal stability was shown to be affected due to the presence of I543V, V732I, S534N, P633Qfs*, D949V variants.

Clustering of data with other populations:

DPYD genotype-based clustering of populations showed three distinct clusters: i) Indians with South Asians and Asians; ii) Africans with African Americans, and iii) Europeans alone.

Clinical validation in pooled Indian data

Pooled data analysis of Indian data showed null association of C29R, I543V and M166V variants with 5-FU/Capecitabine toxicity while V732I, S534N and rs3918290 were associated with 5-FU/Capecitabine toxicity.